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Indian Standard

AERIAL BUNCHED CABLES — FOR WORKING VOLTAGES UP TO AND INCLUDING 1 100 VOLTS — SPECIFICATION

UDC 621·315·211 (621·315·616-96) : 621·396·67

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

April 1995

Price Group 4

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Power Cables Sectional Committee had been approved by the Electrotechnical Division Council.

This standard specifies the requirement of Polyethylene/crosslinked polyethylene insulated cables with aluminium conductors twisted over a central bare/Insulated aluminium alloy messenger wire for use as overhead distribution feeders for rated voltages up to and including 1 100 V.

Many countries are now increasingly using cables with insulated conductor for LT overhead lines. This system is commonly called ABC system (Aerial Bunched Cables).

In the ABC system, the insulated conductors (3 or 4 numbers as required) are twisted around a high strength aluminium alloy bearer wire, which carries the main weight and also serves as the earth-cum-neutral wire. The bearer wire (messenger) may be bare or insulated. The phase conductors are not under tension.

The system is very safe and LT faults are practically eliminated. Consequently, interruptions of supply are minimized and the life of transformers is considerably increased as these are not subjected to frequent faults as happens with the bare conductors. The problem of tree clearance is also minimized.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

AERIAL BUNCHED CABLES — FOR WORKING VOLTAGES UP TO AND INCLUDING 1 100 VOLTS — SPECIFICATION

SECTION 1 GENERAL

1 SCOPE

1.1 This standard covers the requirements of polyethylene/crosslinked polyethylene insulated cables with aluminium conductors twisted over a central bare/insulated aluminium alloy messenger wire for use as overhead distribution feeders.

1.2 The cables covered in this standard are suitable for use on three phase ac (earthed or unearthed) system for rated voltage up to and including 1 100 V.

NOTE — The cables conforming to this standard may be operated continuously at a power frequency voltage 10 percent higher than the rated voltage.

1.3 The cables covered in this standard are suitable for use where the combination of ambient temperature and temperature rise due to load, including temperature on exposure to direct sunlight results in conductor temperature not exceeding the following:

Type of Insulation	*Normal Continuous Operation*	*Short Circuit Operation*
Polyethylene	70°C	160°C
Crosslinked polyethylene	90°C	250°C

2 REFERENCES

2.1 The following Indian Standards are necessary adjuncts to this standard:

IS No.	*Title*
398 (Part 4) : 1979	Aluminium conductors for overhead transmission purposes : Part 4 Aluminium alloy stranded conductors (aluminium-magnesium-silicon type) (*second revision*)
1885 (Part 32) : 1993	Electrotechnical vocabulary : Part 32 Electric cables (*first revision*)
8130 : 1984	Conductors for insulated electric cables and flexible cords (*first revision*)
10418 : 1982	Drums for electric cables
10810 (series)	Methods of test for cables. (The relevant part numbers are given in col 4 of **10.1**)

3 TERMINOLOGY

3.0 For the purpose of this standard the following definitions, in addition to those given in IS 1885 (Part 32) : 1993 shall apply.

3.1 Routine Tests

Tests made by the manufacturers on all finished cable lengths to demonstrate the integrity of the cables.

3.2 Type Test

Tests required to be made before supply on a general commercial basis on a type of cable in order to demonstrate satisfactory performance characteristics to meet the intended application.

NOTE — These tests are of such a nature that after they have been made, they need not be repeated unless changes are made in the cable materials or design which might change the performance characteristics.

3.3 Acceptance Tests

Tests carried out on samples taken from a lot for the purpose of acceptance of the lot.

3.4 Optional Tests

Special tests to be carried out, when required by agreement between the purchaser and the supplier.

3.5 Earthed System

An electric system which fulfills any of the following conditions:

a) The neutral point or mid point connection is earthed in such a manner that even under fault conditions, the maximum voltage that can occur between any conductor and earth does not exceed 80 percent of the nominal system voltage;

b) The neutral point or mid point connection is not earthed but a protective device is installed which automatically cuts out any part of the system which accidentally becomes earthed; or

c) In case of ac system only, the neutral point is earthed through an arc suppression coil with arrangement for isolation within 1 hour of occurrence of the fault provided that total of such periods in a year does not exceed 125 hours.

3.6 Unearthed System

An electric system which does not fulfil the requirements of earthed system (*see* **3.5**).

SECTION 2 MATERIALS

4 CONDUCTOR

4.1 Phase/Street Lighting Conductor

The conductor shall be of H2 or H4 grade aluminium complying with the requirements of IS 8130 : 1984.

4.2 Messenger (Neutral Conductor or Otherwise)

The conductor shall be of heat treated aluminium-magnesium- silicon alloy wires containing approximate 0.5 percent magnesium and approximately 0.5 percent silicon conforming to IS 398 (Part 4) : 1979.

5 INSULATION

5.1 The insulation shall be either of cross-linked polyethylene compound or of polyethylene compound, conforming to requirement given in Tables 1 and 2 respectively.

Table 1 Properties of XLPE Insulation

(*Clause* 5.1)

Sl No.	Tests	Requirements
(1)	(2)	(3)
i)	Tensile strength	12.5 N/mm^2, *Min*
ii)	Elongation at break	200 percent, *Min*
iii)	Ageing in air oven:	
	a) Treatment : Temperature	135 ± 3°C
	Duration	7 days
	b) Tensile strength variation	±25 percent, *Max*
	c) Elongation variation	±25 percent, *Max*
iv)	Hot set:	
	a) Treatment : Temperature	200 ± 3°C
	Time under load	15 min
	Mechanical stress	20 N/cm^2
	b) Elongation under load	175 percent, *Max*
	c) Permanent elongation (set) after cooling	15 percent, *Max*
v)	Shrinkage:	
	a) Treatment : Temperature	130 ± 3°C
	Duration	1 hour
	b) Shrinkage	4 percent, *Max*
vi)	Water absorption (Gravimetric):	
	a) Treatment : Temperature	85 ± 2°C
	Duration	14 days
	b) Water absorbed	1 mg/cm^2, *Max*
vii)	Volume resistivity:	
	a) at 27°C	1 x 10^{13} ohm-cm, *Min*
	b) at 70°C	1 x 10^{11} ohm-cm, *Min*
viii)	Carbon black:	Under consideration
	a) Content	
	b) Dispersion	

Table 2 Properties of PE Insulation
(*Clause* 5.1)

Sl No.	Property	Requirements
i)	Volume resistivity:	
	a) at 27°C	1×10^{14} ohm-cm, *Min*
	b) at 70°C	1×10^{12} ohm-cm, *Min*
ii)	Tensile strength	10 N/mm^2, *Min*
iii)	Elongation at break	300 percent, *Min*
iv)	Melt flow index	3, *Max*
v)	Vicat softening point	85°C, *Min*
vi)	Carbon black:	
	a) Content	Under consideration
	b) Dispersion	Under consideration
vii)	Environmental stress cracking	No cracks shall be visible to naked eye

SECTION 3 CONSTRUCTION

6 CONDUCTOR

6.1 The power/outer insulated neutral/street lighting conductors shall conform to flexibility class 2 of IS 8130 : 1984.

6.2 The messenger (Neutral conductor) or otherwise shall be either stranded circular or compacted circular type and shall have minimum 7 strands. The surface of the conductor shall be smooth.

6.3 A protective barrier may be applied between the conductor and insulation, if required. Such barrier, when used, shall be compatible with insulating material and suitable for the operating temperature of the cable.

6.4 The size of street lighting conductor shall be 16 mm^2.

6.5 The size of messenger conductor and its breaking load shall be as per Table 3.

6.6 There shall be no joints in any wire of the messenger conductor except those made in the base rod or wires before final drawing. The direction of outer layer of wires in messenger conductor shall be right hand.

7 INSULATION

7.1 The conductor (with protective barrier, wherever applied) shall be provided with crosslinked polyethylene or polyethylene insulation applied by extrusion.

Table 3 Size and Requirements of Messenger Conductor
(*Clause* 6.5)

Sl No.	Nominal Cross Sectional Area of Phase Conductor	Messenger Conductor		
		Nominal Cross Sectional Area	Maximum dc Resistance at 20°C	Minimum Breaking Load
	mm^2	mm^2	ohm/km	kN
(1)	(2)	(3)	(4)	(5)
i)	16	25	1.38	7.0
ii)	25	25	1.38	7.0
iii)	35	25	1.38	7.0
iv)	50	35	0.986	9.8
v)	70	50	0.689	14.0
vi)	95	70	0.492	19.7

7.2 Thickness of Insulation

The average thickness of insulation shall be not less than the nominal value (t_i) specified in Table 4.

Table 4 Thickness of Insulation

Nominal Area of Conductor	Nominal Thickness (t_i)
mm^2	mm
(1)	(2)
16	1.2
25	1.2
35	1.2
50	1.5
70	1.5
95	1.5

7.3 Tolerance on Thickness of Insulation

The smallest of measured values of thickness of insulation shall not fall below the nominal value (t_i) specified in Table 4 by more than 0.1 mm + 0.1 (t_i).

7.4 Application of Insulation

The insulation shall be so applied that it fits closely on the conductor (or barrier, if any) and it shall be possible to remove it without damaging the conductor.

7.5 The colour of insulation shall be black.

8 CORE IDENTIFICATION

8.1 The phase conductors shall be provided with one, two or three 'ridges' and outer neutral insulated conductor, if provided, shall have four 'ridges' as shown in Fig. 1 for quick identification. The street lighting conductor and messenger conductor (if insulated) shall not have any identification mark.

8.2 Identification by other means, as agreed between the supplier and purchaseer, is also permissible.

9 ASSEMBLY (LAYING UP)

9.1 Three insulated phase conductors, one insulated neutral conductor (if required) and a street lighting conductor (if required) shall be twisted around the bare (or insulated) as required messenger conductor without fillers with a lay not exceeding 35 times the diameter of the insulated phase conductor.

9.2 The direction of lay shall be right hand.

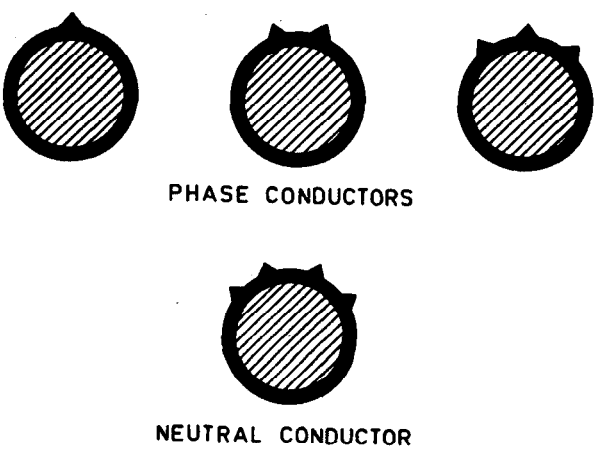

FIG. 1 CORE IDENTIFICATION

SECTION 4 TESTS

10 CLASSIFICATION OF TESTS

10.1 Type Tests

The following shall constitute type tests:

Sl No.	*Tests*	*Requirements*	*Test Method (Ref to Part No. of IS 10810)*
(1)	(2)	(3)	(4)
a)	Tests on phase/street light conductor:		
	i) Tensile test	*See* IS 8130 : 1984	2
	ii) Wrapping test	do	3
	iii) Resistance test	do	5
b)	Test on messenger conductor:		
	i) Breaking load	*See* Table 3	2
	ii) Elongation test	*See* 11.3	11.3
	iii) Resistance test	*See* Table 3	5
c)	Physical tests for XLPE Insulation:		
	i) Tensile strength and elongation at break	*See* Table 1	7
	ii) Ageing in air oven	do	11
	iii) Hot set test	do	30
	iv) Shrinkage test	do	12
	v) Water absorption (Gravimetric)	do	33
	vi) Carbon black:	Under consideration	
	1) Content		
	2) Dispersion		
d)	Physical test for PE insulation:		
	i) Tensile strength and elongation at break	*See* Table 2	7
	ii) Melt flow index	do	23
	iii) Carbon black:		
	1) Content	do	32
	2) Dispersion	do	Under consideration
	iv) Vicat softening point	do	22
	v) Environmental stress cracking	do	29
e)	Test for thickness insulation	*See* Table 4	8
f)	Insulation resistance (Volume resistivity) test	*See* Table 1 and 2	43
g)	High voltage test	*See* 11.2	45

10.2 Acceptance Test

The following shall constitute acceptance tests:

a) Tensile test (for phase/street light conductor),

b) Wrapping test (for phase/street light conductor),

c) Breaking load test for messenger conductor,

d) Elongation test for messenger conductor,

e) Conductor resistance test,

f) Test for thickness of insulation,

g) Tensile strength and elongation at break test,

h) Hot set test for XLPE insulation,

j) Insulation resistance test, and

k) High voltage test.

10.2.1 A recommended sampling plan for acceptance tests is given in Annex A.

10.3 Routine Tests

The following shall constitute routine tests:

a) Conductor resistance test, and

b) High voltage test.

10.4 Optional Test

The following shall constitute optional test:

Bending test.

11 DETAILS OF TEST

11.1 General

Unless otherwise stated in this standard, the tests shall be carried out in accordance with appropriate parts of IS 10810 taking into account additional information given in this standard.

11.2 High Voltage Test

High voltage test at room temperature (Type, Acceptance and Routine Test):

The cable shall withstand a voltage of 3 kV ac (rms) at a frequency of 40 to 60 Hz or a dc voltage of 7.2 kV between conductors for 5 minutes.

11.3 Elongation Test on Messenger Conductor

Test shall be carried out on wires before stranding. The specimen shall be straightened by hand and an original gauge length of 200 mm shall be marked on the wire. A tensile load shall be applied by means of a suitable tensile testing machine. The load shall be applied gradually and the rate of separation of the jaws of the testing machine shall be not less than 25 mm per minute and not greater than 100 mm per minute. The elongation shall be measured after the fractured ends have been fitted together. If the fracture occurs outside the gauge marks, or within 25 mm of either mark, and the required elongation is not obtained, the test shall be disregarded and another test made.

The elongation of the specimen shall be not less than 4 percent on a gauge length of 200 mm.

11.4 Bending Test on the Completed Cable

The test shall be performed on a sample of complete cable. The sample shall be bent around a test mandrel at room temperature for at least one complete turn. It shall then be unwound and the process shall be repeated after turning the sample around its axis by 180 degrees. The cycle of these operations shall then be repeated twice. The diameter of the mandrel shall be :

$$10\,(D + d)$$

where

D = actual diameter of the cable, for example, minimum circumscribing diameter, mm; and

d = actual diameter of the phase conductor, mm.

Requirement : No cracks visible to the naked eye are allowed.

SECTION 5 IDENTIFICATION, PACKING AND MARKING

12 IDENTIFICATION

12.1 Manufacturers, Identification

The manufacturer shall be identified throughout the length of the cable by means of a tape bearing the manufacturers name or trade-mark and year of manufacture. In case the above method cannot be employed, or if the purchaser so desires, colour identification threads in accordance with a scheme to be approved by the Bureau of Indian Standards (BIS) shall be employed.

12.2 Cable Identification

Cables with cross-linked polyethylene insulation shall be identified throughout the length of the cable by the legend 'XLPE 90' and year of manufacture by a printed tape. No identification is required for polyethylene insulation.

NOTES

1 Single tape bearing manufacturer's name or trade-mark and the cable identification, that is, XLPE 90, if provided, shall be acceptable against the requirement of **12.1** and **12.2**.

2 Identification required in **12.1** and **12.2** shall be given on any one of the insulated conductor of cables.

13 PACKING AND MARKING

13.1 The cable shall be wound on a drum (*see* IS 10418 : 1982) of suitable size and packed. The ends of the cable shall be sealed by non-hygroscopic material.

13.2 The cable shall carry the following information

either stencilled on the drum or contained in a label attached to it:

a) Reference to this Indian Standard, for example, Ref IS 14255 ;
b) Manufacturer's name or trade-mark;
c) Type of cable and voltage grade;
d) Nominal cross-sectional area of phase conductor;
e) Length of cable on the drum;
f) Number of lengths on the drum (if more than one);
g) Direction of rotation of drum (by means of an arrow);
h) Gross mass;
j) Country of manufacture; and
k) Year of manufacture.

13.2.1 The cable (drum or label) may also be marked with the Standard Mark.

NOTE — The use of the Standard Mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

SECTION 6 ACCESSORIES FOR AERIAL BUNCHED CABLES

14 The accessories which are in common use for aerial bunched cables are under consideration.

ANNEX A

(*Clause* 10.2.1)

SAMPLING OF CABLES

A-1 LOT

A-1.1 In any consignment the cables of the same size manufactured under essentially similar conditions of production shall be grouped together to constitute a lot.

A-2 SCALE OF SAMPLING

A-2.1 Samples shall be taken and tested from each lot for ascertaining the conformity of the lot to the requirements of the specification.

A-2.2 The number of drums (*n*) to be selected from the lot of drums *(N)* of consignment of cables shall be in accordance with col 2 and 1 of Table 5 respectively. These samples shall be taken at random.

A-2.2.1 In order to ensure the randomness of selection, random number tables shall be used (*see* IS 4905 : 1968 Methods of random sampling).

A-3 NUMBER OF TESTS AND CRITERION FOR CONFORMITY

A-3.1 Suitable length of test sample shall be taken from each of the drums selected. These test samples shall be subjected to each of the acceptance tests (*see* **10.2**). A test sample is called defective if it fails in any of the acceptance tests. If the number of defectives is less than or equal to the corresponding permissible number (*a*) given in col 3 of Table 5 the lot shall be declared as conforming to the requirements of acceptance test, otherwise not.

Table 5 Number of Drums to be Selected for Sampling and Permissible Number of Defectives

Number of Drums in the Lot	Number of Drums to be Taken as Sample	Permissible Number of Defectives
(1)	(2)	(3)
N	*n*	a
Up to 50	2	0
51 to 100	5	0
101 " 300	13	0
301 " 500	20	1
501 and above	32	2

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